

AI-Powered S&P 500 Constituent Change Monitor



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PROBLEM STATEMENT

Changes to the constituents of the S&P 500 are important for investors, researchers, and index-tracking fund managers.

However, identifying S&P 500 constituent changes as soon as they are announced is a time-consuming process, as it requires continuous monitoring of a large volume of news releases, most of which are unrelated to S&P 500 index changes. Furthermore, the information is presented in unstructured text, making it difficult for traditional automation tools to accurately identify and extract relevant updates.

This project addresses these challenges by leveraging Microsoft Power Automate and AI capabilities to automate the monitoring and documentation workflow.

SOLUTION OVERVIEW

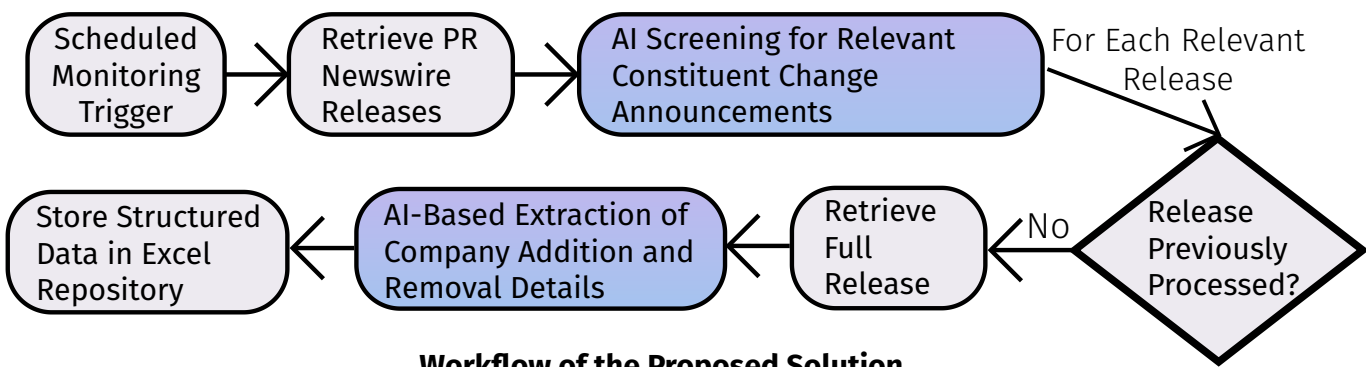
An AI-powered S&P 500 Constituent Change Monitor was developed using Microsoft Power Automate, AI Builder, PR Newswire*, Excel Online, and OneDrive.

The solution automatically monitors announcements published by S&P Dow Jones Indices through PR Newswire based on a configurable schedule. Monitoring frequency is configurable for intraday, hourly, or daily execution.

AI is integrated at two key stages of the workflow:

- Announcement classification to identify constituent change events from newly published releases.
- Information extraction to transform unstructured article content into structured data.

Power Automate orchestrates the end-to-end workflow, including article retrieval, AI processing, duplicate detection, and data recording. Leveraging Excel Online and OneDrive enables fully cloud-based and unattended execution.



Workflow of the Proposed Solution

USE of AI

AI Prompt 1: Announcement Classification

The first AI prompt classifies newly published PR Newswire announcements to identify genuine S&P 500 constituent membership changes. Traditional keyword-based filtering is insufficient, as references to the S&P 500 do not necessarily indicate additions, deletions, or replacements within the index. AI-based contextual classification is therefore used to distinguish actual constituent change events from unrelated announcements and updates involving other indices.

For each relevant announcement, the prompt returns the publication date and time, headline, and relative URL in a consistent JSON structure. This filtering stage reduces unnecessary processing and ensures that only relevant announcements are forwarded to the second AI prompt.

AI Prompt 2: Structured Information Extraction

The second AI prompt retrieves and analyses the full content of announcements identified by the first AI prompt, transforming unstructured article content into structured data. Rule-based extraction alone cannot reliably process constituent change announcements because the required information lacks a standardized presentation format and may be intermixed with updates relating to other S&P indices. AI-driven contextual analysis is therefore used to identify, interpret, and extract only S&P 500 constituent changes.

Using contextual analysis, the prompt isolates S&P 500 constituent changes and extracts key attributes, including action type, company name, ticker symbol, effective date, and GICS sector. The model also synthesizes a concise, evidence-based summary of the stated reason for each change rather than simply reproducing source text. Results are returned in a standardized JSON format, enabling automated processing and storage in Excel.

IMPACT & VALUE

By automating the monitoring and analysis of PR Newswire releases, the solution eliminates the need for continuous manual review, enables timely identification of S&P 500 constituent changes, and transforms unstructured announcements into structured, actionable data.

Data Source

*Data Source: PR Newswire. S&P Dow Jones Indices News Releases.
<https://www.prnewswire.com/news/s%26p-dow-jones-indices/>

REFLECTION

Key Learnings

This project demonstrated how AI can create value by transforming unstructured information into structured, actionable data. Although platforms such as Bloomberg Terminal already provide access to S&P 500 constituent changes, the framework developed in this project offers a reusable approach for automating the monitoring and extraction of information from unstructured sources.

Challenges Encountered

A key challenge was identifying a suitable source for constituent change announcements. The original plan was to retrieve announcements directly from the S&P Dow Jones Indices Media Center, but anti-scraping measures restricted automated access. Several alternative sources were subsequently evaluated based on reliability, timeliness, and compatibility with automated retrieval. As PR Newswire serves as an official distribution channel for many S&P Dow Jones Indices press releases, it was ultimately selected as the most suitable source.

Potential Future Enhancements

Future enhancements include integrating stock price data before and after constituent changes to analyse their impact on stock performance.